Self-testing quantum random number generator

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QCRYPT 2015 Tokyo

Joint work with



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Charles Ci Wen Lim



Tommaso Lunghi



Anthony Martin



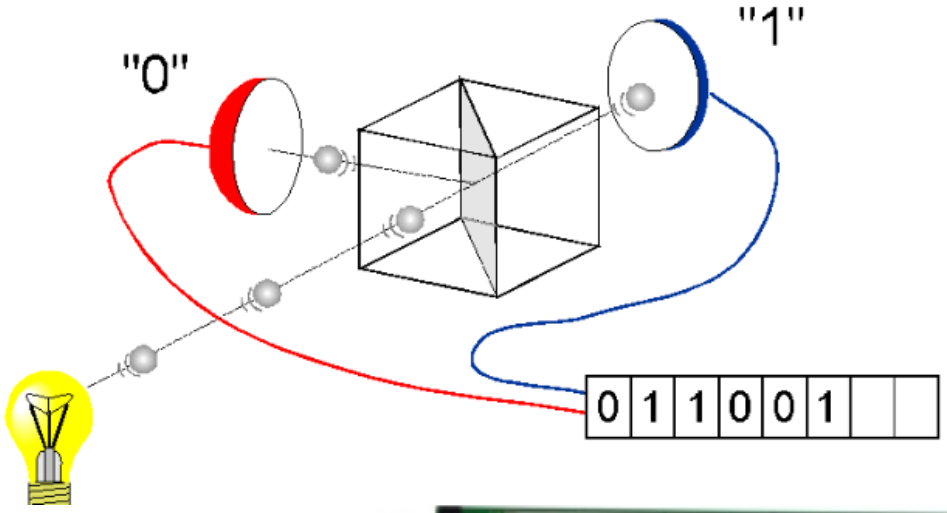
Quentin Lavigne



Hugo Zbinden

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Group of Applied Physics

Practical QRNG (trusted device)







Rate: MHz → GHz

Device-independent (black-box) QRNG

Vol 464|15 April 2010|doi:10.1038/nature09008 nature

LETTERS

Random numbers certified by Bell's theorem

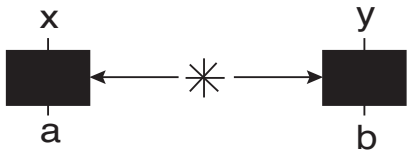
S. Pironio[1,2]*, A. Acín[3,4]*, S. Massar[1]*, A. Boyer de la Giroday[5], D. N. Matsukevich[6], P. Maunz[6], S. Olmschenk[6], D. Hayes[6], L. Luo[6], T. A. Manning[6] & C. Monroe[6]

Rate: 42 bits / month

Experimental Requirements

Security / level of paranoia

Fully Device Independent



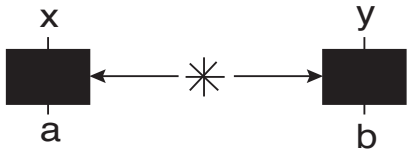


Device Dependent

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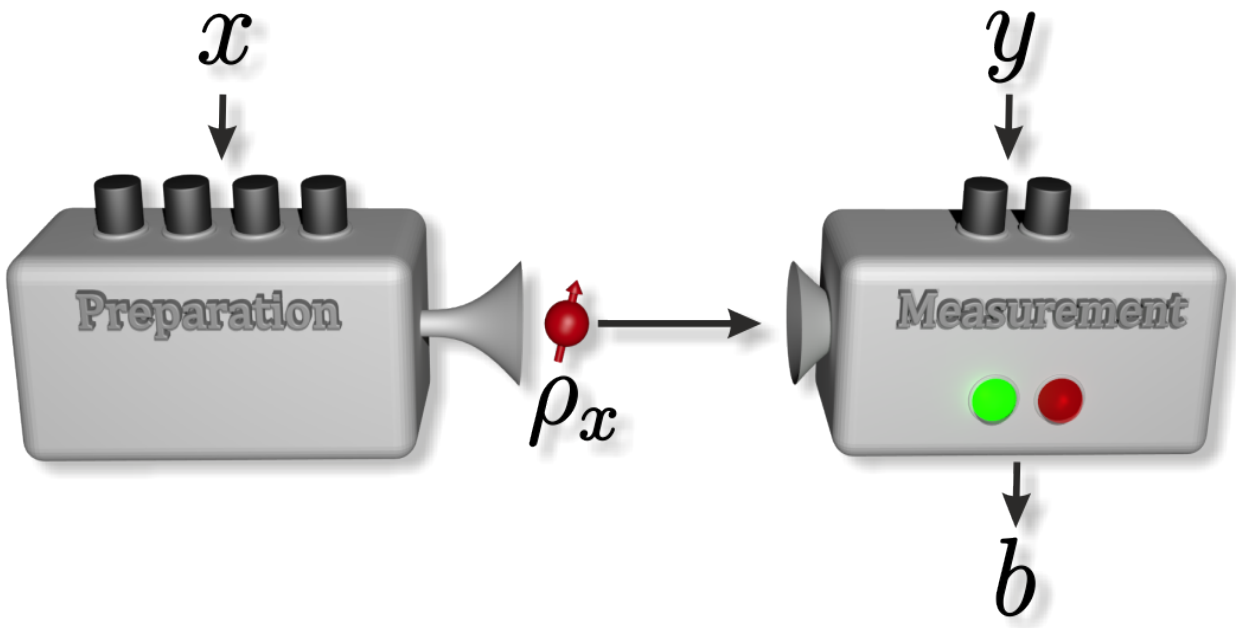
THIS WORK

non malicious devices subject to imperfections

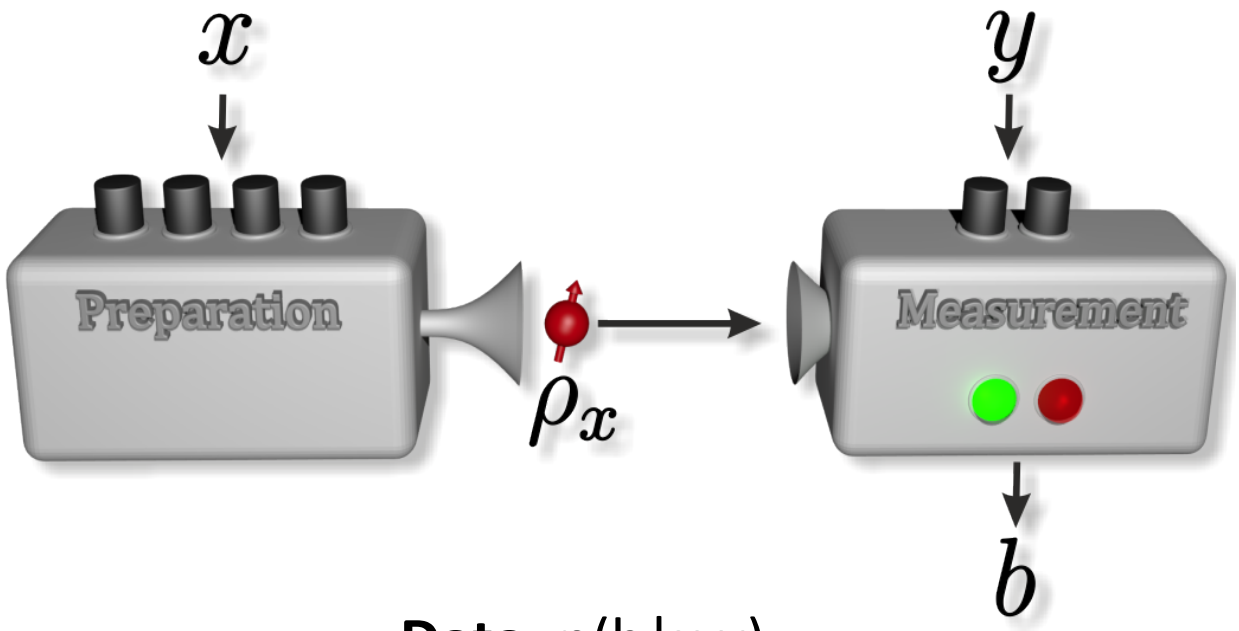


Device Dependent

Scenario: Non malicious but error-prone devices

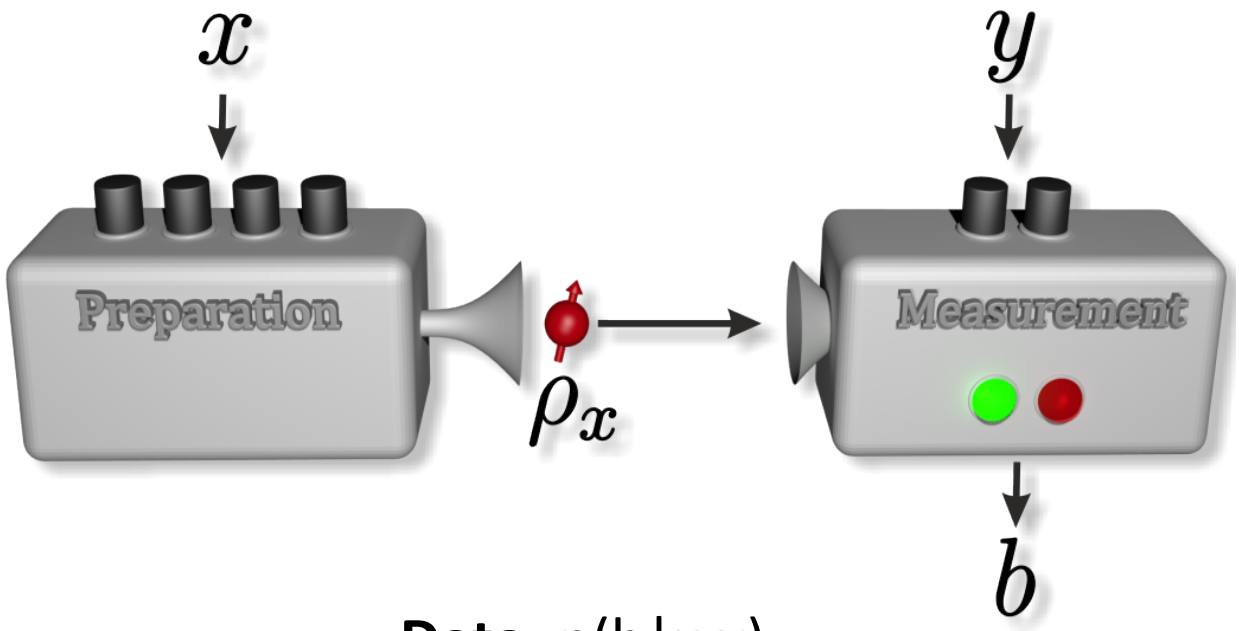


Scenario: Non malicious but error-prone devices



Data: p(b|x,y)

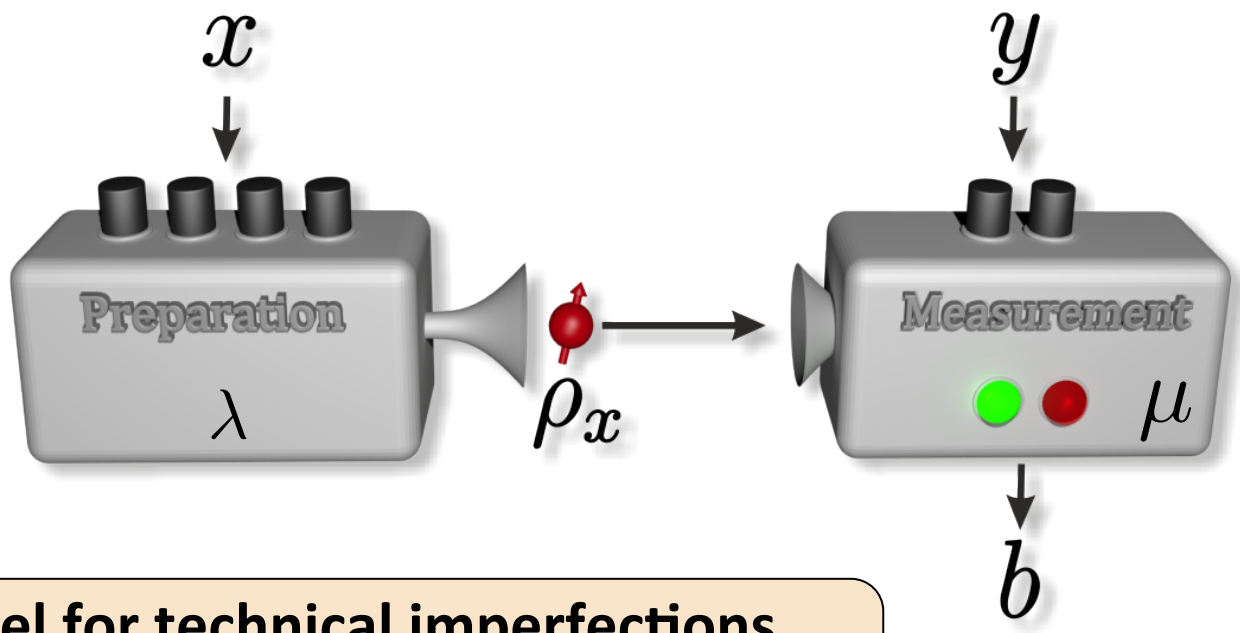
Scenario: Non malicious but error-prone devices



Data: p(b|x,y)

Difficulty: Source of apparent of randomness in b

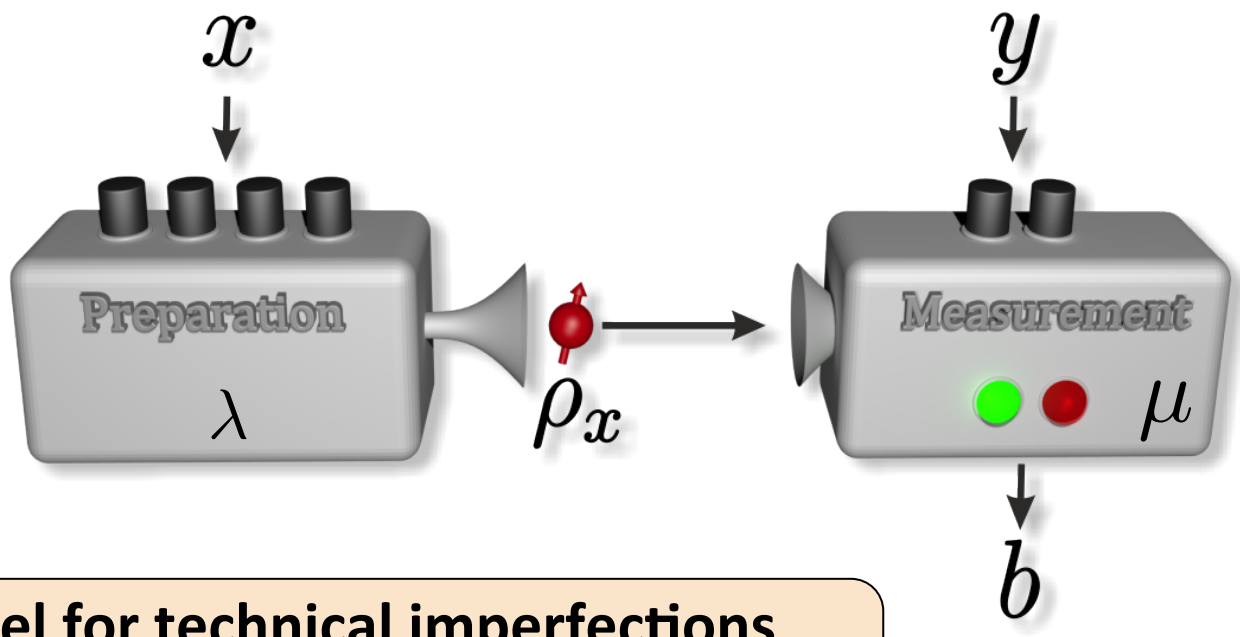
Q RANDOMNESS ? or **TECHNICAL NOISE ?**



Model for technical imperfections

Internal states of the devices λ μ

$$p(b|xy) = \sum_{\lambda,\mu} q_\lambda r_\mu p(b|xy\lambda\mu)$$

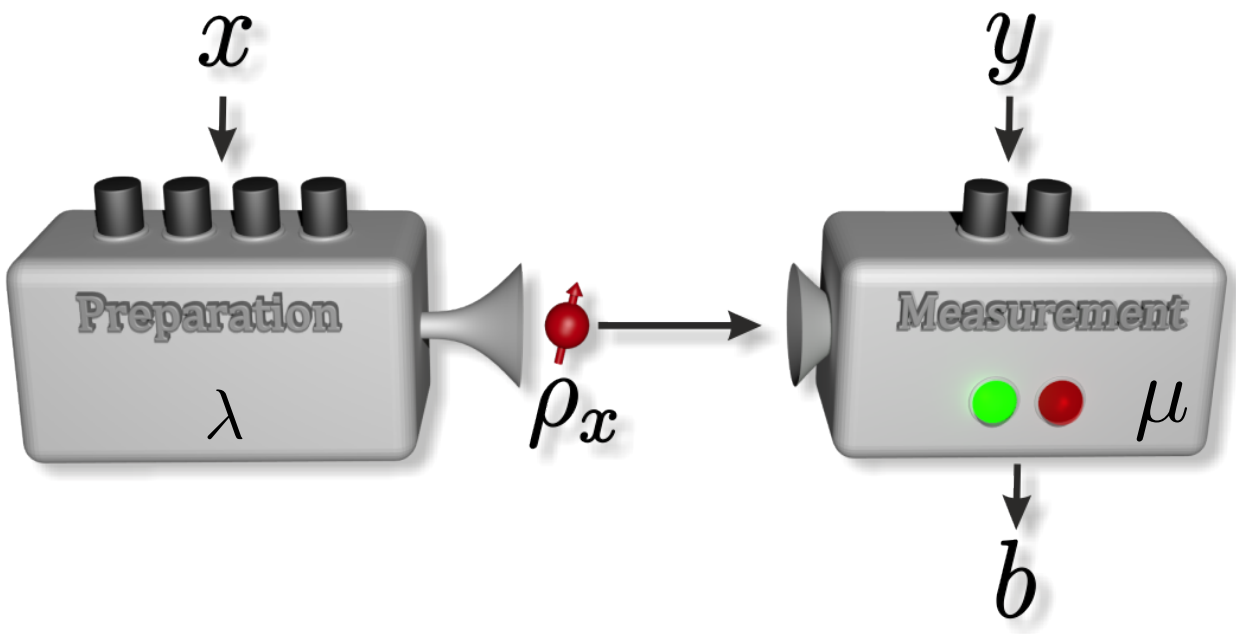


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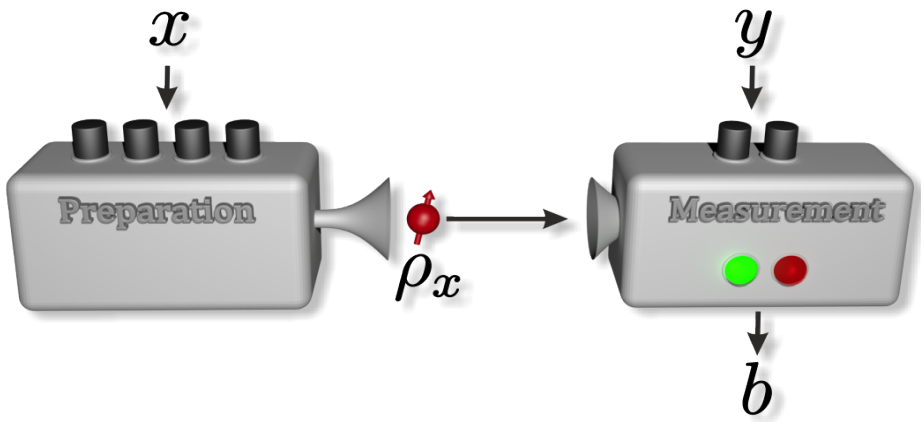
Guessing Probability $p^g_{xy} = \sum_{\lambda\mu} q_\lambda r_\mu \max_b p(b|xy\lambda\mu)$



Assumptions

- Uncorrelated devices
- Inputs x and y generated independently from devices
- x encoded in qubit
- Memory-less devices (after-pulsing tolerated)

Witness



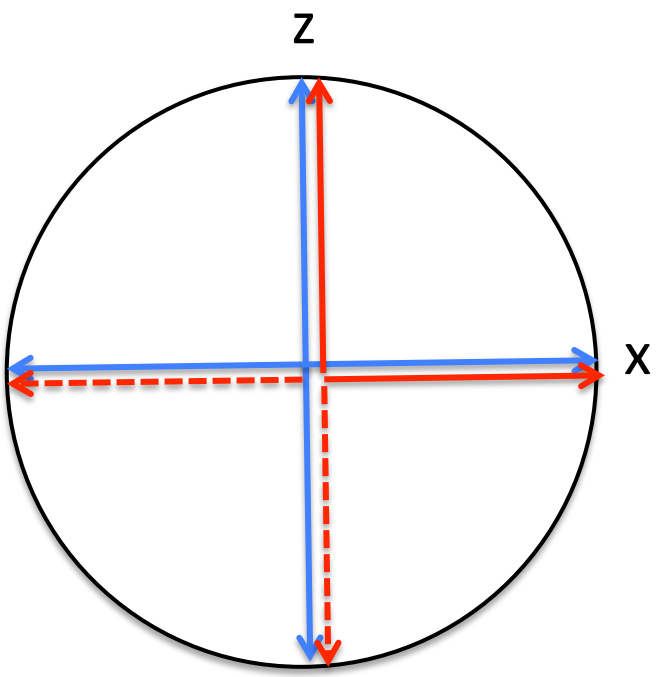
$$W = |\det \begin{pmatrix} p(0|0,0) - p(0|1,0) & p(0|2,0) - p(0|3,0) \\ p(0|0,1) - p(0|1,1) & p(0|2,1) - p(0|3,1) \end{pmatrix}|$$

Classical bits → W = 0

→ Qubits can do better: W > 0

Bowles et. al. PRL 2014

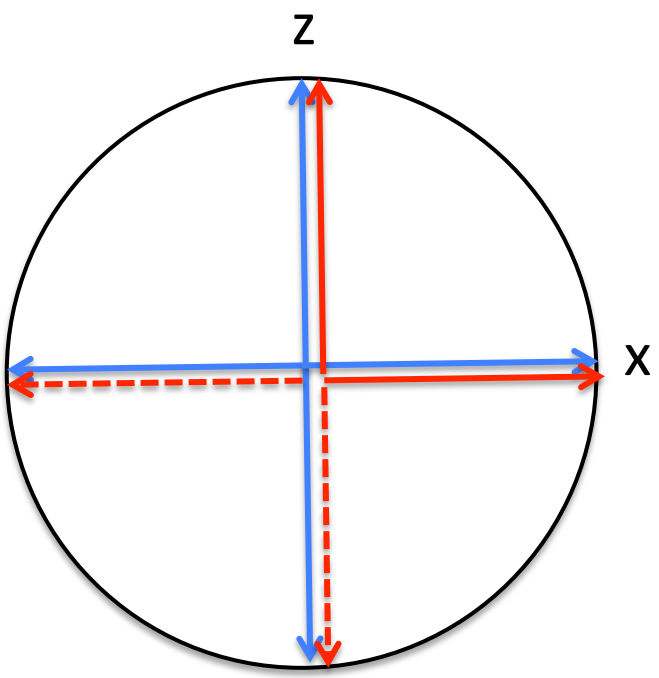
Qubits



BB84 states and measurements

W = det (id) = 1

Qubits



BB84 states and measurements

$W = \det(id) = 1$

Effect of imperfections

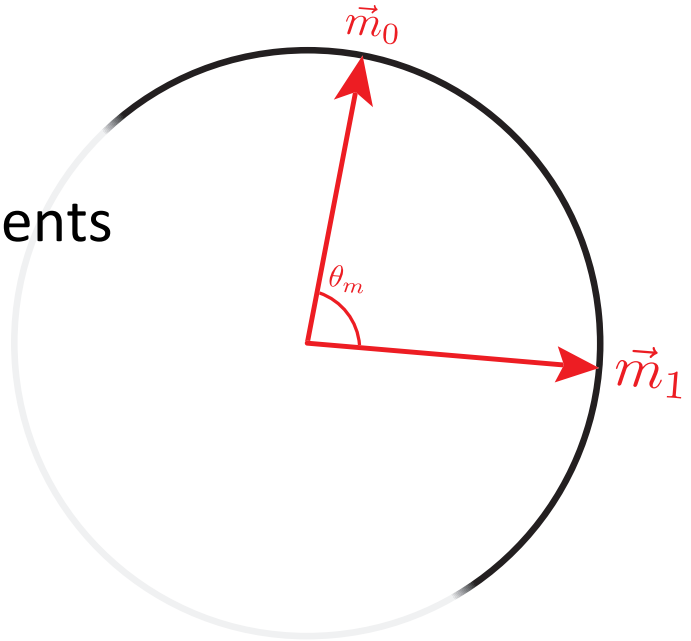
Visibility V ; efficiency η $\quad W = (V\,\eta)^2$

Can tolerate **arbitrarily low** efficiency and visibility

Quantifying randomness

W quantifies **incompatiliby of** measurements

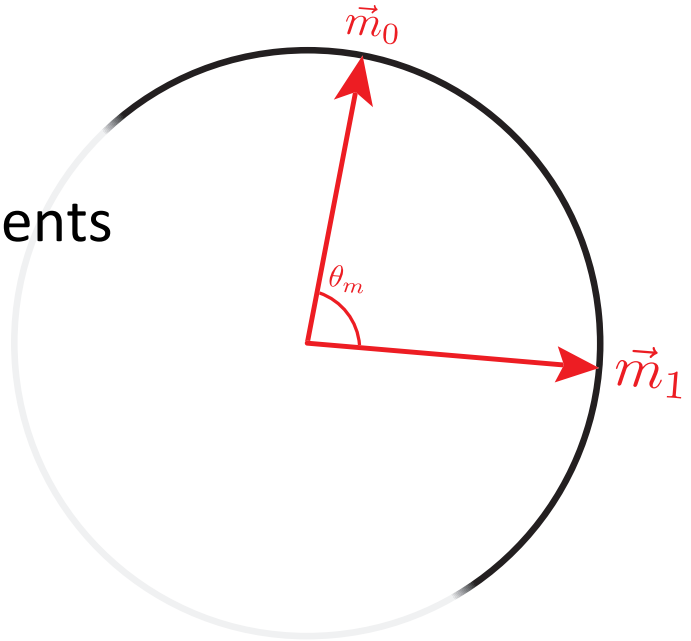
$$\sin(\theta_m) \geq W$$



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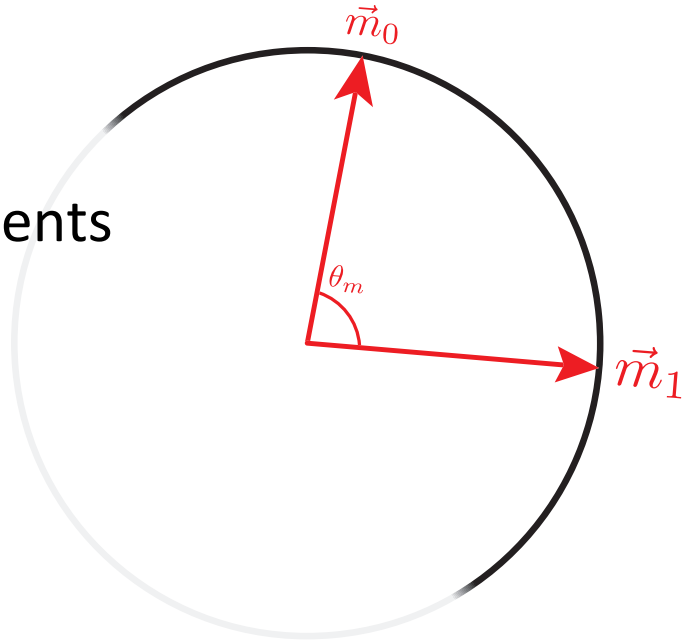


W > 0 → Measurements are incompatible → Randomness

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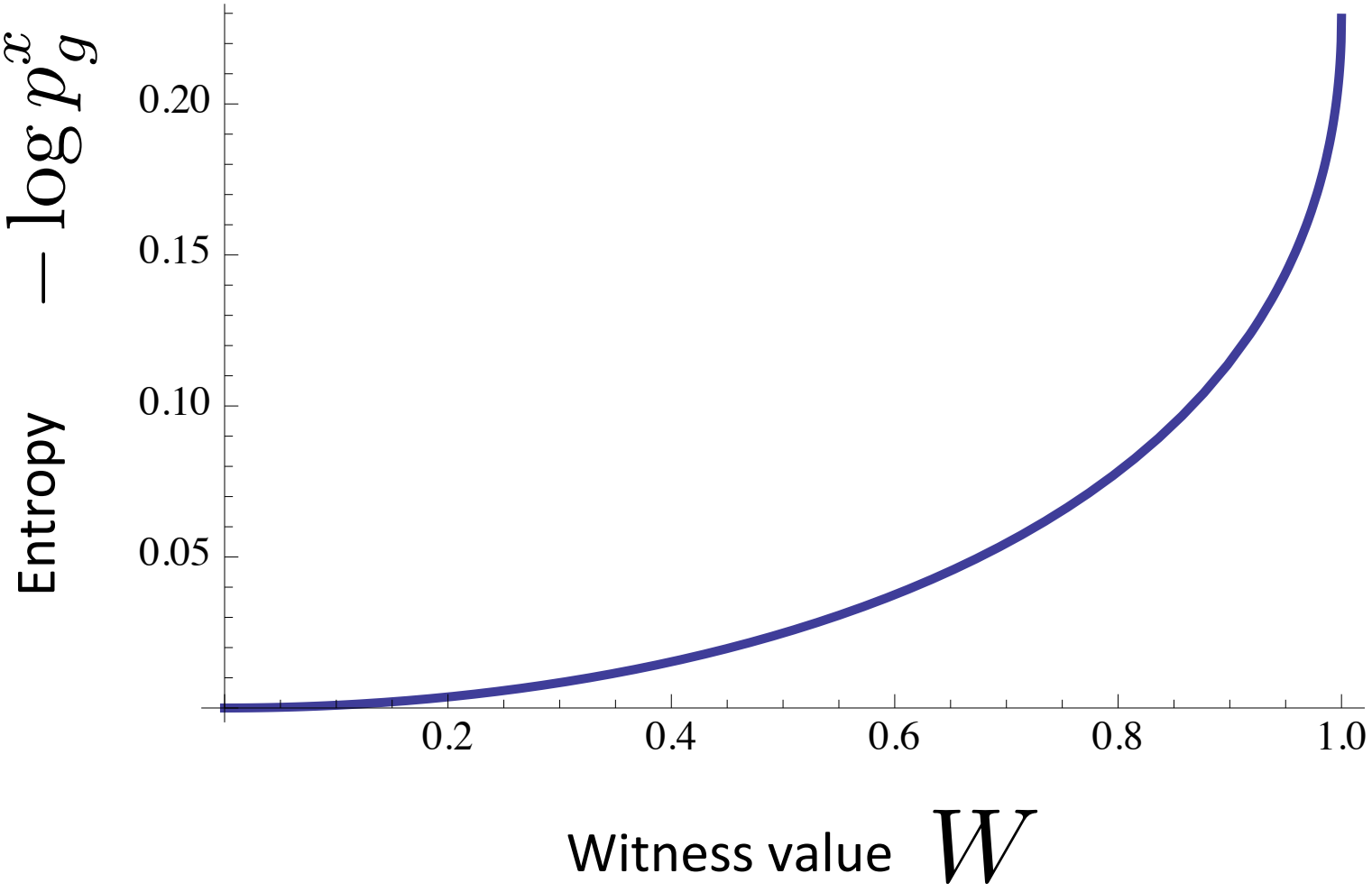
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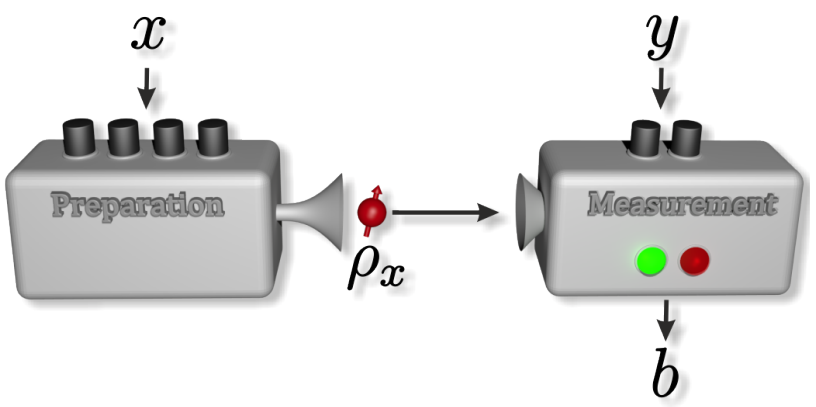
Quantitatively: $p_g^x \leq \cos^2(\theta_m/2) \leq \frac{1}{2}\left(1+\sqrt{\frac{1+\sqrt{1-W^2}}{2}}\right)$

Bounds on randomness



Self-testing QRNG - Protocol

Experiment



Data: {x, y, b}

Entropy estimation

a. Probability: $p(b|x,y)$

b. Witness:

$$W = \begin{vmatrix} p(1|0,0) - p(1|1,0) & p(1|2,0) - p(1|3,0) \\ p(1|0,1) - p(1|1,1) & p(1|2,1) - p(1|3,1) \end{vmatrix}$$

c. Entropy:

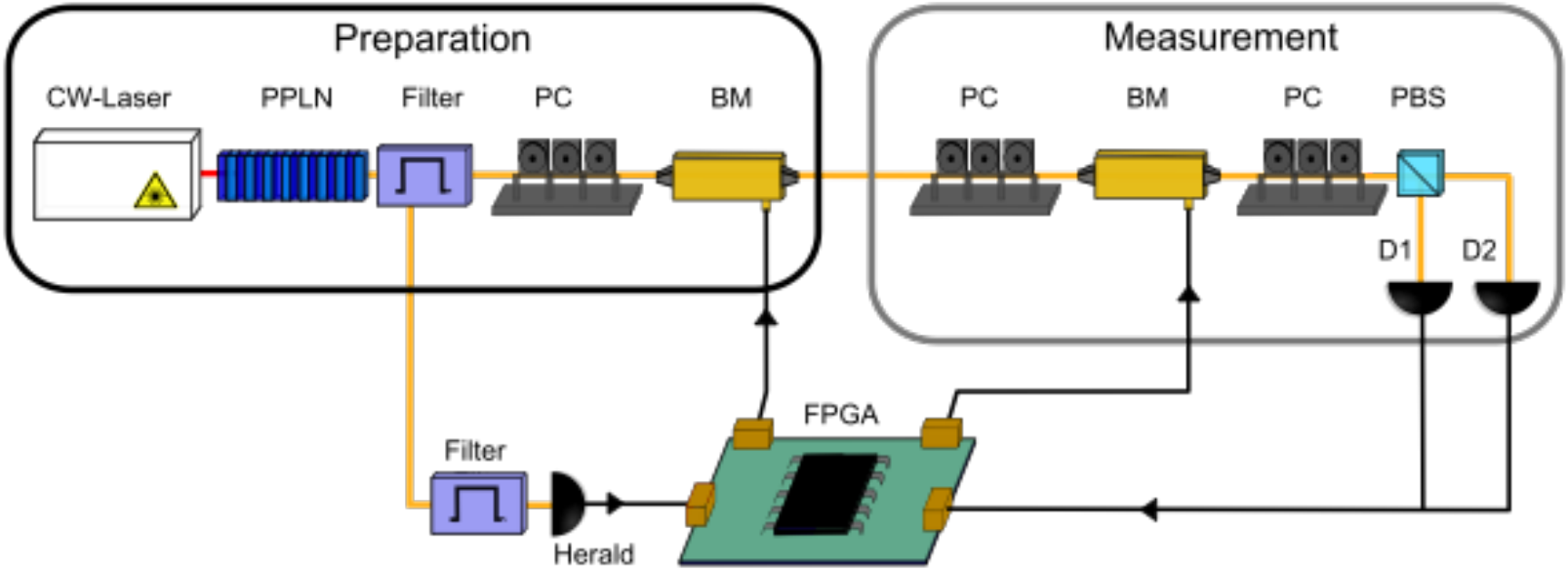
$$H_{min} \geq -\log_2 \left(\frac{1}{2} \left(1 + \sqrt{\frac{1+\sqrt{1-W^2}}{2}} \right) \right)$$

Random bit extraction

Classical randomness extraction
Toeplitz matrix

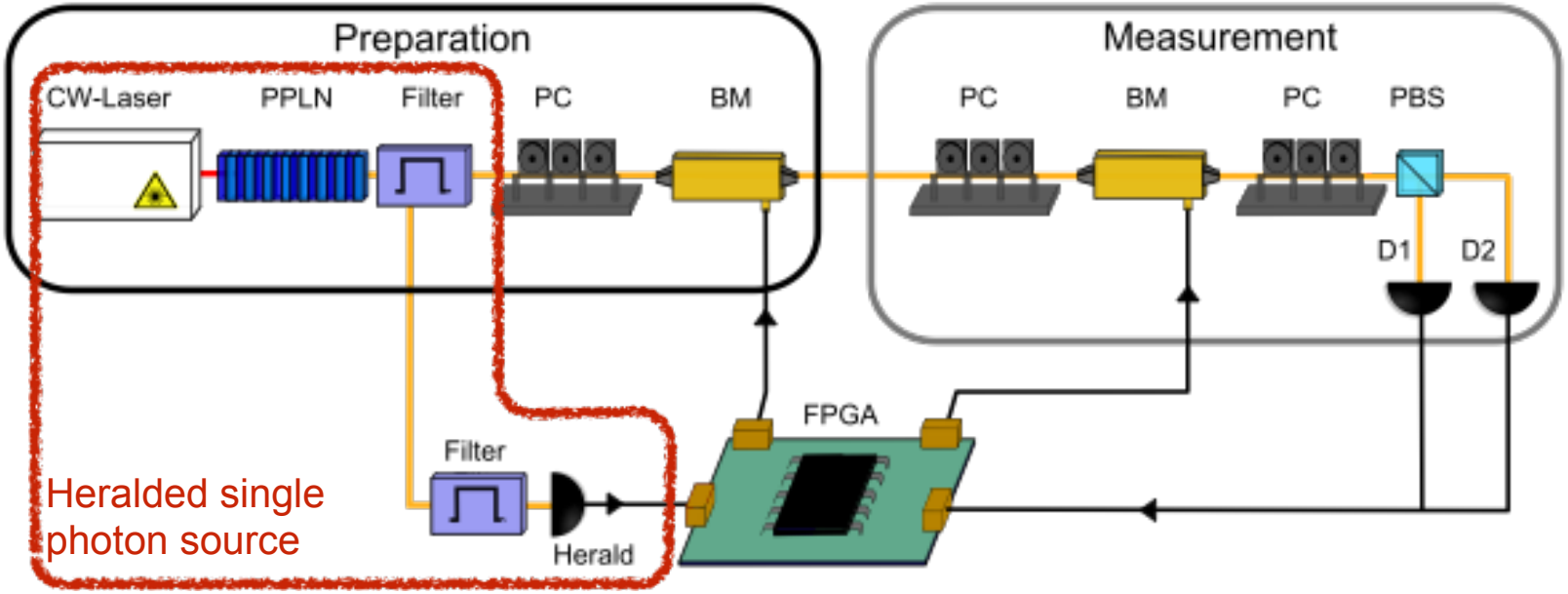
Experimental setup

Qubits = polarisation of single photons



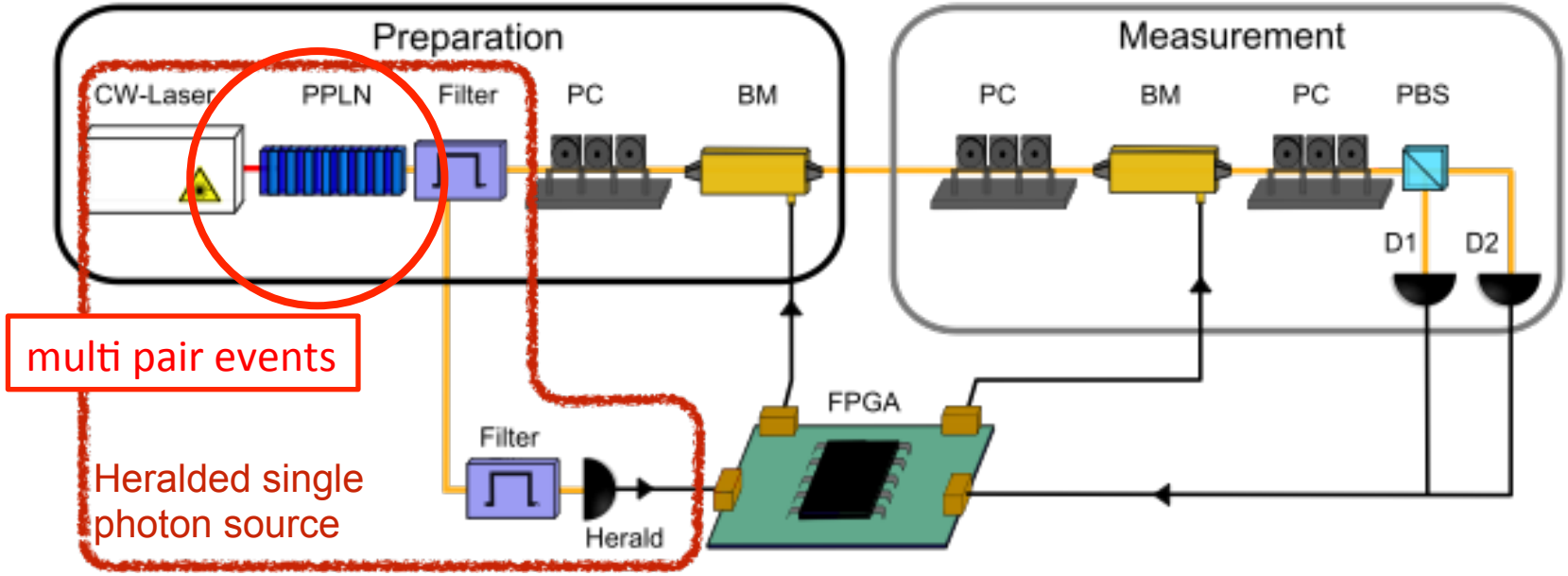
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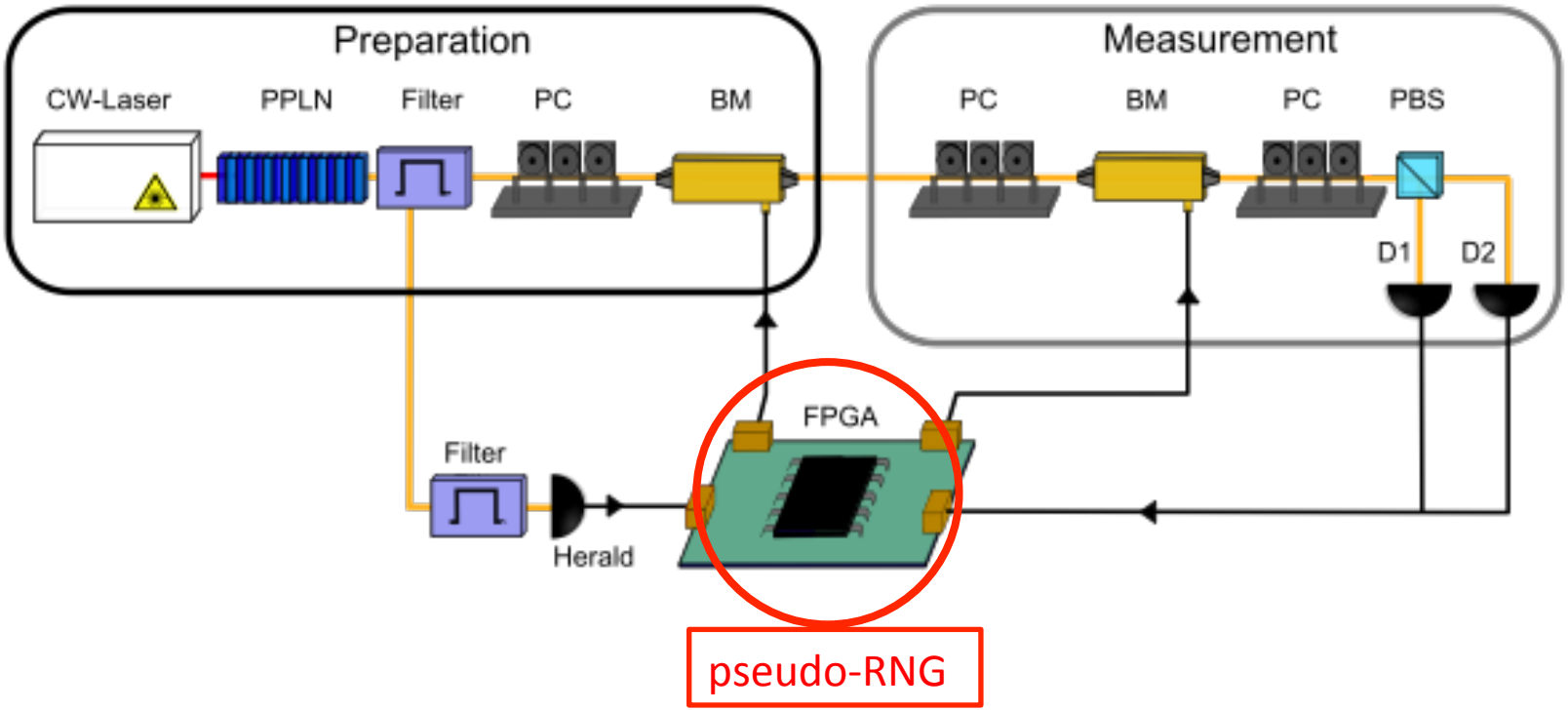
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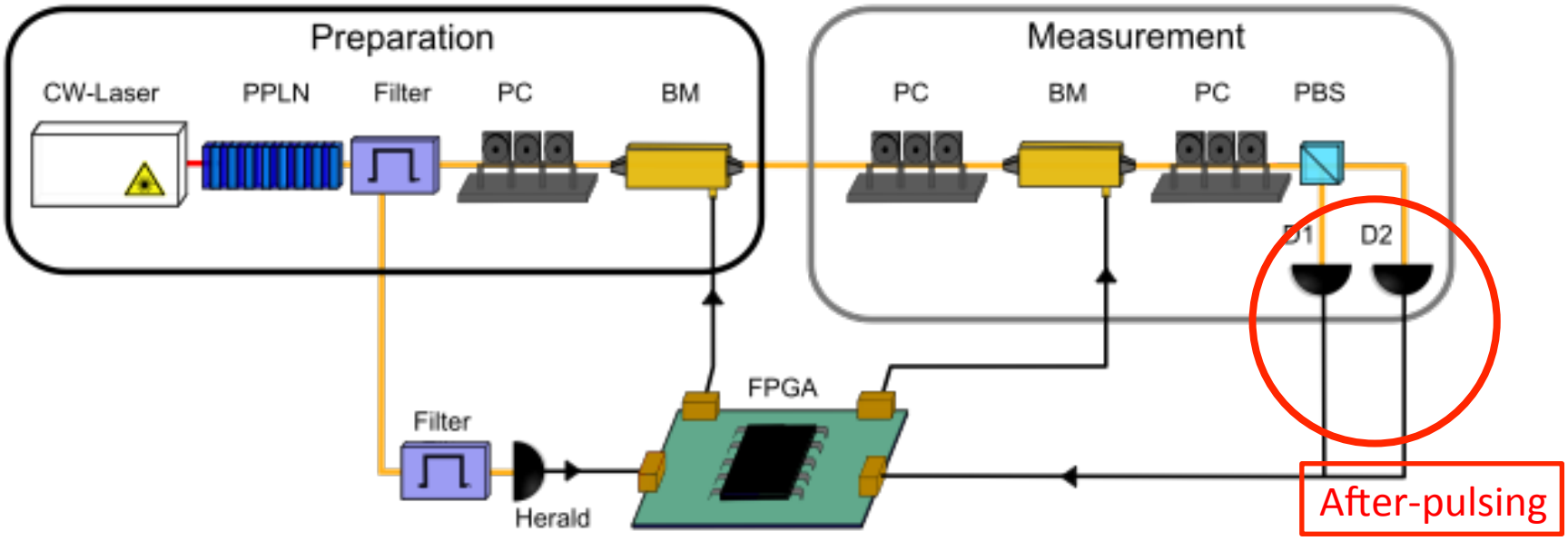
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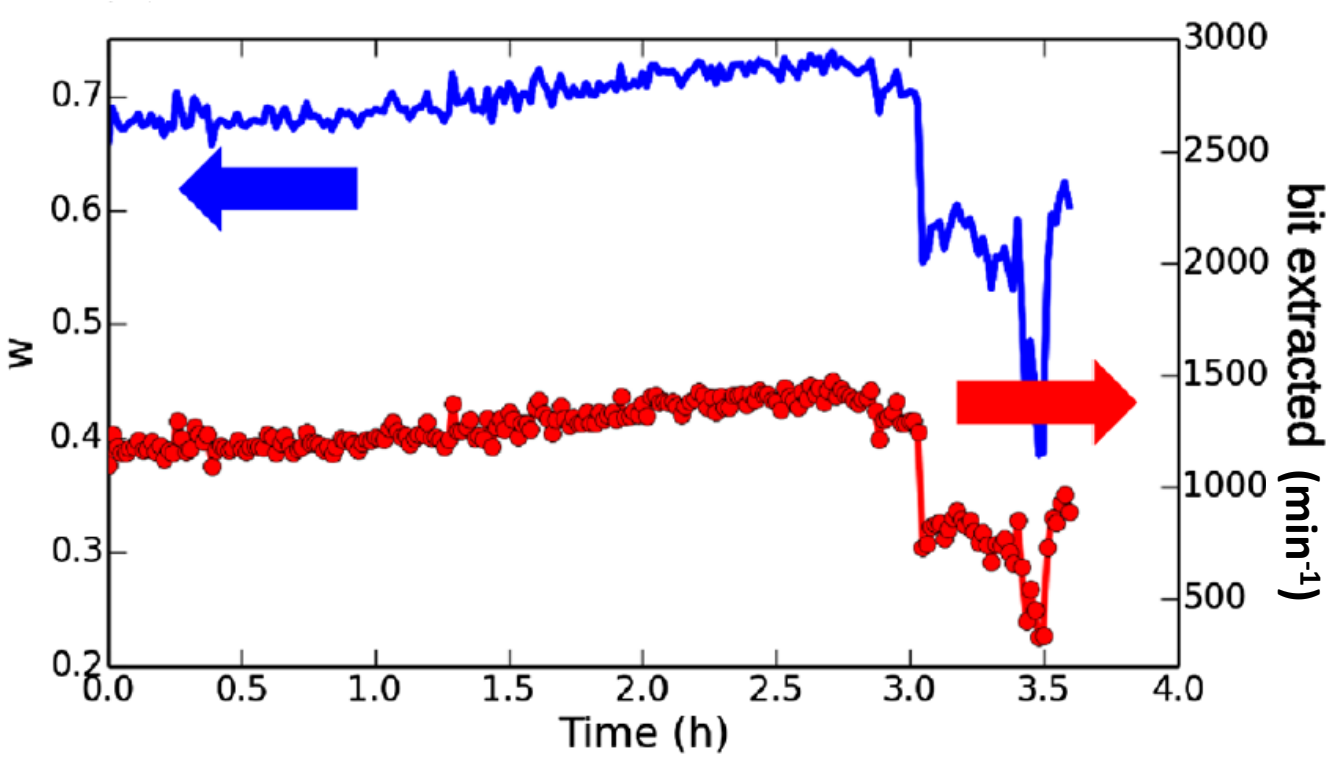


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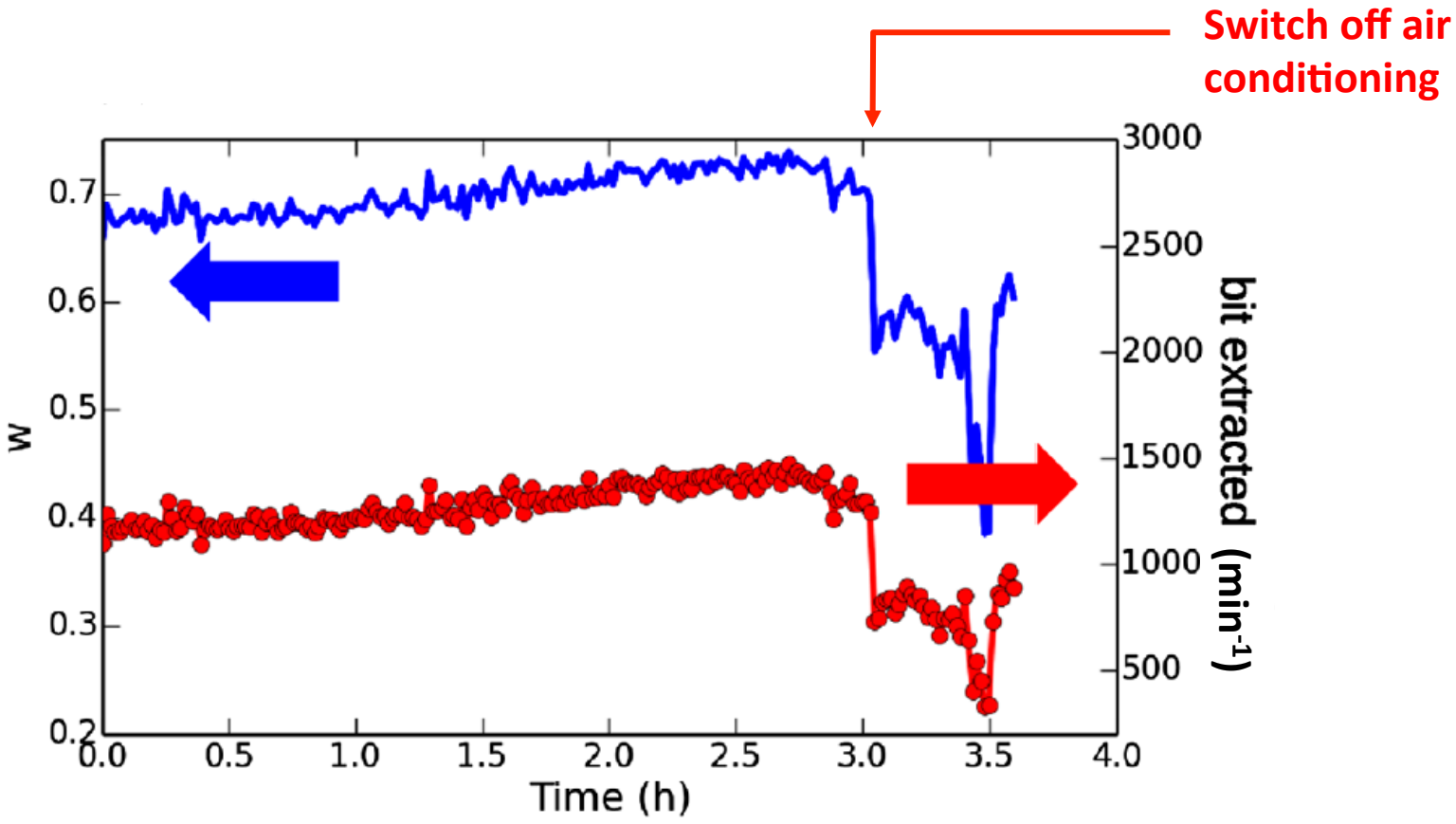
Results



With fair sampling - 24 bits/sec (W=0.76)

Without fair sampling - 6 x 10^{-5} bits/sec (W=1.5 x 10^{-4})

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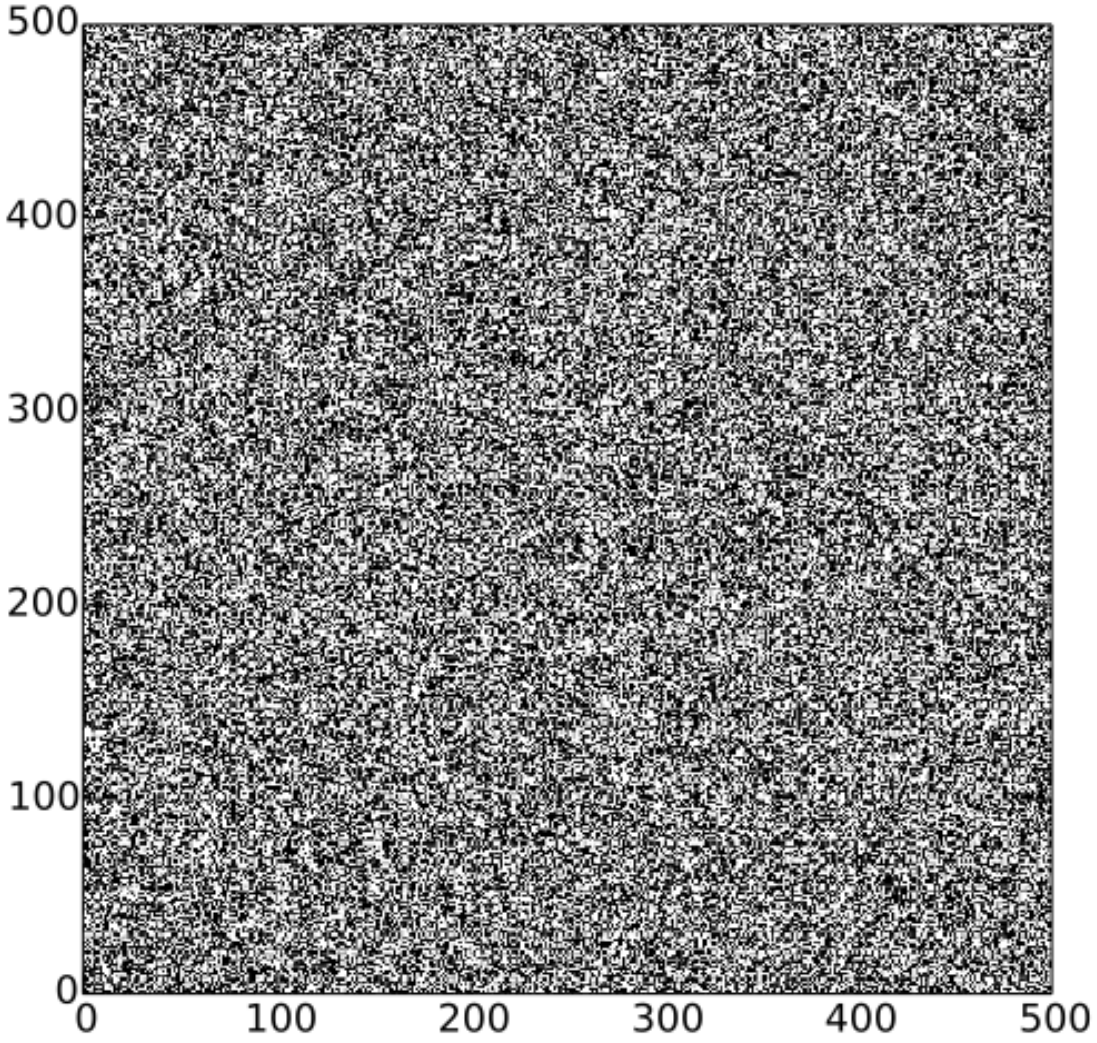
Conclusion

- Approach to QRNG intermediate **between device-independent and fully trusted**

  see also Vallone et al. PRA'14, Canas et al. arxiv'14
- Error-prone but non malicious devices
- **Real-time entropy estimation**
- Robust to noise and detector inefficiency
- Few general assumptions:
  - Qubit preparations
  - Independent devices
  - No memory

Lunghi et. al. PRL **114**, 150501 (2015)

RNG Test



Satisfied NIST test

